

1 Q1 Tool Safari & Creative Mini Projects (15 pts)

1.1 Artifacts & Tools

Image (optical illusion). HF Space *AP123/IllusionDiffusion*¹ with base *SG161222/Realistic_Vision_V5.1_noVAE* (SD 1.5 finetune) and ControlNet *monster-labs/control_v1p_sd15_qrcode_monster*³.

Audio (kids-show BGM). HF Space *nsaintsever/music-generation*⁴ using *facebook/musicgen-small*⁵.

1.2 Prompts, Runs, and Files

I logged every run in `prompts.jsonl` (image: 4 prompt families $\times$ two illusion strengths; audio: 7 prompts at 10s). I keep the selected outputs in `outputs.jsonl`. Screenshots and artifacts are in `artifacts/`. Exact prompt strings and reproducibility notes are in the Appendix.

1.3 Creativity Twist & Setup

Twist (image): create an optical illusion where a forest scene (owl + panther) reads as a hidden woman’s face (Lena pattern as control).

Setup: seed=22; guidance=7.5; sampler=Euler; ControlNet start=0, end=1; `illusion_strength` $\in \{0.8, 1.5\}$; concise negatives. I kept 3 labeled variants for evidence: *success* (P1@1.5), *borderline* (P2@0.8), *fail* (P3@0.8).

1.4 Qualitative Comparison (compact)

Aspect	Image (IllusionDiffusion)	Audio (MusicGen)
Fluency	Clean renders; illusion readable when alignment is right.	Clean 10s clips; no artifacts.
Control	<i>Text & negative prompts</i> ; seed; CFG/guidance scale; sampler (Euler); <code>illusion_strength</code> ; ControlNet guidance start/end; upscaler strength; <i>control image</i> (Lena)	Prompt steers instruments/tone; UI mainly exposes duration (no BPM/seed/temperature).
Style	<i>Realistic</i> (more foliage detail) vs. <i>oil-paint</i> (flatter tones).	Early-2000s bed: soft piano/bells, diatonic, loop-friendly.
Errors	Literal portrait; odd facial parts on scenery.	Occasionally too energetic vs. “whisper-quiet” ask.
Speed/UX	~10s/image after GPU alloc; easy knobs.	~30–40 s/clip; simple but limited controls.

¹<https://huggingface.co/spaces/AP123/IllusionDiffusion>

²https://huggingface.co/SG161222/Realistic_Vision_V5.1_noVAE

³https://huggingface.co/monster-labs/control_v1p_sd15_qrcode_monster

⁴<https://huggingface.co/spaces/nsaintsever/music-generation>

⁵<https://huggingface.co/facebook/musicgen-small>

One failure I inspected. Prompt 3: “Quiet forest at dawn, few trees only... minimal background.” (neg: “Multiple trees, multiple animals”). At strength 1.5 it tended to a literal face; at 0.8 it produced an “almost face” with odd features—useful structure was missing, so the illusion broke.

1.5 Foundation Models & Architectural Difference

IllusionDiffusion. Latent *diffusion* (SD 1.5) with CLIP text encoder and a U-Net denoiser; *ControlNet* injects image-structured conditioning while leaving base weights mostly fixed.

MusicGen. *Autoregressive* decoder-only Transformer over EnCodec audio tokens; generates token sequences from text then decodes to waveform.

Key contrast. Diffusion (iterative denoising in latent space with spatial conditioning) vs. autoregression (token-by-token sequence modeling); CLIP+image control vs. text-only conditioning on audio codes.

2 Q2 Evaluation: Automated & Human (20 pts)

I came across **intrinsic** and **extrinsic** evaluation. Intrinsic uses only the model’s probabilities on gold data (no generation). Extrinsic uses generated outputs and an external comparator or rule (e.g., a reference, a feature encoder, or test cases).

2.1 Automated metrics

2.1.1 Perplexity — *intrinsic* (text; lower is better)

Intuition. Average branching factor of the model’s next-token distribution on the reference.

Formulas (equivalent).

$$\begin{aligned} \text{PP} &= \exp\left(-\frac{1}{N} \sum_{t=1}^N \log p_{\theta}(x_t \mid x_{<t})\right) \quad (\text{exp-log form}) \\ &= \left(\prod_{t=1}^N p_{\theta}(x_t \mid x_{<t})\right)^{-1/N} = P(x_{1:N})^{-1/N} \quad (\text{multiplication form}). \end{aligned}$$

Example (bigram LM; teacher-forced). Sentence (10 tokens): the quick brown fox jumps over the lazy dog again.

Toy bigram probabilities:

$$p(\text{the} \mid \text{BOS}) = \frac{1}{5}, \quad p(\text{quick} \mid \text{the}) = \frac{1}{2}, \quad p(\text{brown} \mid \text{quick}) = \frac{1}{2}, \quad p(\text{fox} \mid \text{brown}) = \frac{1}{5}, \quad p(\text{jumps} \mid \text{fox}) = \frac{1}{2}, \quad p(\text{over} \mid \text{jumps}) = \frac{1}{2},$$

Product = $\frac{1}{38400}$, hence

$$\text{PP} = (38400)^{1/10} \approx \mathbf{2.874}.$$

2.1.2 BLEU — *extrinsic* (text; higher is better)

Intuition. Phrase-overlap pressure: rewards reuse of short n -grams from the reference in the right order; penalizes being too short and over-repeating beyond the reference.

Formula (BLEU- N).

$$\text{BLEU}_N = \text{BP} \cdot \exp\left(\frac{1}{N} \sum_{n=1}^N \log p_n\right), \quad \text{BP} = \begin{cases} 1, & c > r, \\ \exp(1 - r/c), & c \leq r, \end{cases}$$

where p_n are *clipped* n -gram precisions, c is candidate length, and r is reference length.

Example (BLEU-2). Reference: a small gray cat sleeps ($r=5$). Candidate: small gray cat sleep ($c=4$).

Unigrams: matches = 3 $\Rightarrow p_1 = 3/4$. Bigrams: matches = 2 $\Rightarrow p_2 = 2/3$.

Brevity penalty: $\text{BP} = e^{1-5/4} = e^{-0.25} \approx 0.7788$. Geometric mean: $\sqrt{(3/4)(2/3)} = \sqrt{1/2} \approx 0.7071$.

$$\Rightarrow \text{BLEU}_2 \approx 0.7788 \times 0.7071 \approx \mathbf{0.55}.$$

2.1.3 FID — *extrinsic* (images; lower is better)

Intuition. Feature-cloud distance between real and generated images using a fixed encoder (Inception pool3). Two parts: (i) center shift (means) and (ii) spread/shape mismatch (covariances). If feature clouds overlap, FID ≈ 0 .

Formula (Fréchet distance in feature space).

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r^{1/2}\Sigma_g\Sigma_r^{1/2})^{1/2}\right),$$

with (μ_r, Σ_r) from real features and (μ_g, Σ_g) from generated features.

Example (2-D; diagonal covariances). Real: $\mu_r = (0, 0)$, $\Sigma_r = I$. Gen: $\mu_g = (1, 2)$, $\Sigma_g = \text{diag}(4, \frac{1}{4})$.

Mean term = $\|(1, 2)\|^2 = 5$. Cov term = $(1 + 4 - 2 \cdot 2) + (1 + \frac{1}{4} - 2 \cdot \frac{1}{2}) = 1 + 0.25 = 1.25$.

$$\Rightarrow \text{FID} = 5 + 1.25 = \mathbf{6.25}.$$

2. Human evaluation: Describe two approaches (e.g., pairwise preference, rubric ratings). Discuss trade-offs: cost, bias, inter-rater reliability, instruction drift.

2.2 Human evaluation

2.2.1 Pairwise preference (A/B; ranking)

Description. For each prompt/item, I show two (or a small set of) system outputs; the judge chooses the better one (or provides a ranking). Aggregate choices into a global ordering or scores.

Model (Bradley–Terry).

$$P(i \text{ beats } j) = \frac{e^{s_i}}{e^{s_i} + e^{s_j}},$$

Estimating the latent scores $\{s_i\}$ from win/loss data yields a system ranking.

Design/computation. Use randomized order and blinding. Sampling can be: (i) full pairwise per prompt (complete graph), or (ii) adaptive tournaments (Elo/TrueSkill/dueling-bandit scheduling) to cut redundant comparisons.

Trade-offs.

- **Cost.** Naïve full comparisons scale $\mathcal{O}(M^2)$ per prompt; adaptive scheduling reduces toward $\mathcal{O}(M \log M)$. More judgments per item than single-pass scoring.
- **Bias.** Minimizes scale-use bias (no numeric scales). Residual risks: position/order and presentation bias $\Rightarrow$ randomize sides, balance positions, blind model IDs.
- **IRR.** Typically strong per judgment because the task is discrete. Report agreement rate on winners, Kendall’s τ between judges’ induced rankings, or consistency of Bradley–Terry fits.
- **Instruction drift.** Lower: “which is better?” is concrete. Track with periodic calibration pairs and spot audits.

2.2.2 Rubric ratings (absolute/anchored scales; attribute rubrics)

Description. Judges score each output on a defined scale (e.g., 1–5, 0–100), optionally across attributes (e.g., adequacy, fluency, coherence, harmlessness). Anchors (reference examples) and brief training define what each score means. Scores can be normalized per rater (e.g., z-scores) before averaging.

Design/computation. Specify the rubric, provide positive/negative exemplars, and fix anchors over the study. For multi-attribute rubrics, report per-attribute means and optionally a weighted composite.

Trade-offs.

- **Cost.** Single pass per item ($\mathcal{O}(M)$), easy to batch; yields attribute-level diagnostics.
- **Bias.** Vulnerable to leniency/severity, central-tendency, and anchoring effects. Mitigations: explicit anchors, exemplars, short calibration, per-rater normalization.
- **IRR.** Often lower without tight rubrics. Use ICC (interval/continuous) or Krippendorff’s α (ordinal/interval); increase raters per item if needed.
- **Instruction drift.** Higher risk over time as judges reinterpret scales. Mitigate with fixed anchors, versioned guidelines, refresher calibration, and gold items seeded across sessions.

2.3 3) Benchmarks (Q2.3)

[3]

2.3.1 ARC (AI2 Reasoning Challenge)

What it measures

- **Focus on reasoning:** needs multi-sentence understanding and comparisons, not just fact lookup
- **Multiple-choice format:** 4 options per question
- **Science focus:** grade-school science (roughly 3rd–9th grade)
- **Shot format:** supports 0-shot and few-shot settings (e.g., 25-shot)

Why use ARC

- **Shifts focus to reasoning** instead of recall
- **Multiple difficulty levels** within science
- **Standardized evaluation** across models on the same set

Sample item

Which of the following architectures in generative AI is more recent?

- A) GAN
 - B) VAE
 - C) Autoregressive Transformer
 - D) Diffusion
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2.3.2 GSM-8K (Grade School Math 8K)

What it measures

- **Focus on multi-step reasoning:** read a word problem, break it into steps, do the math
- **Large and diverse dataset:** 8.5K high-quality problems
- **Focus on accuracy:** final output is a **single number** (exact-match scoring)
- **Shot format:** works with 0-shot or few-shot worked examples

Why use GSM-8K

- **Practical application:** mirrors school and day-to-day arithmetic
- **Tests core skills:** planning, calculation, and checking
- **Standardized evaluation:** same problems for fair model comparison

Sample item

Fifteen years ago, Ram's age was half of his brother Shyam's age. Today, their age difference is 6 years. How old are Ram and Shyam now? (Answer as two integers: Ram, Shyam.)

2.4 What current evaluations miss (using HELM)

1) Calibration (knowing when the model is unsure)

Many leaderboards show accuracy but not whether confidence matches correctness. HELM reports calibration next to accuracy and finds noticeable gaps even for strong models.

Example metric: **ECE-10** (lower is better). [1]

2) Faithfulness (grounded, non-hallucinatory answers)

Fluent text can still make unsupported claims. HELM adds targeted tests (e.g., disinformation) and often relies on human judgment when automatic factuality scores fall short.

Example metric: **human-judged claim support** (supported / contradicted / unsure) on long answers. [1]

3) Long-context reasoning

Many tasks need information spread over very long inputs. HELM introduced a Long-Context track to measure how performance changes as context grows.

Example metric: **accuracy vs. context length** (e.g., at 32k, 100k tokens), plus speed/efficiency at those lengths. [2]

4) Multilingual and pragmatic coverage

HELM’s core is mostly English, and it flags missing languages and varieties as a gap. Typical evaluations miss dialects, register, and pragmatics.

Example metric: **group-wise performance gaps** across languages/dialects (slice accuracy or perplexity; disparity indices). [1]

5) Safety beyond a single toxicity score

Safety has many facets (toxicity, bias, misinformation). HELM breaks this out by scenario and shows that rates vary by prompt context.

Example metric: **% toxic generations** on datasets like RealToxicityPrompts/BOLD (stratified by prompt type), plus **bias** scores where relevant. [1]

3 Q3 Measure What I Made (10 pts)

3.1 Metric (choice & why)

Objective. I want images that *suggest* a woman’s face via alignment with a control pattern while avoiding literal portraits and high-clutter textures.

Score. I use an edge-overlap score (F_2) with recall emphasis, a continuous portrait penalty, and an edge-density penalty:

$$\text{IllusionScore} = F_2 - 0.9 \cdot \text{portrait_pen} - 0.25 \cdot \text{EDP}.$$

I apply 1 px dilation tolerance. Implementation details and the notebook cell are in the Appendix.

3.2 Application to my Q1 images

Data. I generated 8 images (seed 22) using four prompts at two illusion strengths. I reference images by filename and `prompt_id` from `prompts.jsonl`:

- q1-image-p1-s0p8, q1-image-p1-s1p5 (“Forest at dawn break ... panther crouched ... owl on another”) $\Rightarrow$ illusion_1.webp, illusion_2.webp
- q1-image-p2-s0p8, q1-image-p2-s1p5 (adds “Optical illusion, hidden woman’s face, clear edges.”) $\Rightarrow$ illusion_3.webp, illusion_4.webp
- q1-image-p3-s0p8, q1-image-p3-s1p5 (“few trees only ... minimal background”) $\Rightarrow$ illusion_6.webp, illusion_5.webp
- q1-image-p4-s0p8, q1-image-p4-s1p5 (adds “precise edges, gentle oil-paint look”) $\Rightarrow$ illusion_7.webp, illusion_8.webp

The full table is in `eval.csv` (Appendix).

Results (rank by IllusionScore). illusion_2.webp (q1-image-p1-s1p5) > illusion_8.webp (q1-image-p4-s1p5) > illusion_7.webp (q1-image-p4-s0p8) > illusion_1.webp > illusion_4.webp > illusion_3.webp > illusion_6.webp > illusion_5.webp.

What boosted vs. hurt (prompt wording effects with concrete cases).

- **Stronger imprint improved recall without a face.** illusion_2.webp (q1-image-p1-s1p5, strength 1.5) scored **0.386** = F_2 0.414, portrait 0.000, EDP 0.110. Raising strength from illusion_1.webp (0.8) *increased recall* (0.589 $\rightarrow$ 0.614) and slightly *reduced EDP* (0.136 $\rightarrow$ 0.110) while keeping portrait_pen at 0.
- **“gentle oil-paint look” reduced clutter but increased portrait-ness.** Oil-paint (Prompt 4) made edges cleaner and tones flatter: illusion_8.webp (q1-image-p4-s1p5) scored **0.377** = F_2 0.456, portrait 0.074, EDP 0.045. Compared to realistic, EDP $\downarrow$ and F_2 often $\uparrow$, but portrait_pen increases slightly as smooth facial surfaces appear.
- **Explicit “hidden woman’s face” helped alignment but triggered penalties at high strength.** illusion_4.webp (q1-image-p2-s1p5) beat its low-strength pair on F_2 (0.377 vs. 0.292) and EDP (0.025 vs. 0.128) *but* incurred portrait_pen 0.099, pulling its overall score down.
- **“few trees only ... minimal background” over-simplified the scene.** illusion_5.webp/illusion_6.webp (Prompt 3) achieved EDP $\approx$ 0, yet F_2 stayed low ($\sim$ 0.28) due to under-coverage of the control edges; portrait_pen $\sim$ 0.08–0.09 when the face became explicit.

Style observation. Oil-paint $\Rightarrow$ EDP $\downarrow$, F_2 $\uparrow$, portrait_pen $\uparrow$ (small); realistic $\Rightarrow$ higher EDP from foliage but often portrait_pen=0.

3.3 Improvement plan (LoRA) with risks

Plan. I will train a small LoRA (rank 8, $\alpha \approx 16$, lr 1×10^{-4} , 300–800 steps) on a licensed, curated mini-set of optical-illusion scenes (implied faces via silhouettes/branches; exclude literal portraits), with light augmentation. I will adopt a simple gate on a held-out set: $\Delta \text{IllusionScore} \geq 0.03$, portrait_pen $\leq$ baseline+0.02, EDP ≤ 0.12 (early stop otherwise). At inference I will keep LoRA weight 0.6–0.8 with negatives like “no portrait/human face” and composition hints (off-center).

Risks & mitigations.

- **Portrait drift.** Mitigate by excluding portrait-heavy images, monitoring validation Illusion-Score (includes penalty), capping LoRA weight.
- **Style lock-in.** Mitigate via balanced (realistic+oil-paint) data, low-rank LoRA/few steps, and style-sliced eval gates.

Notes

I place the scorer cell and formula details in the Appendix. I include `eval.csv` (all 8 rows), the 8 images, and all exact prompt strings in the Appendix for verification.

References

- [1] Percy Liang et al. Holistic Evaluation of Language Models (HELM), 2022. Available: [Link](#)
- [2] Stanford CRFM. HELM Long-Context track (latest results). Available: [Link](#)
- [3] “Ultimate Guide to LLM Benchmarks video” YouTube. Available: [Link](#)

Appendix

A. Reproducibility (settings)

Image tool: HF Space *AP123/IllusionDiffusion*, base *SG161222/Realistic_Vision_V5.1_noVAE*, ControlNet *monster-labs/control_v1p_sd15_qrcode_monster*.

Control: artifacts/samples/test.png.

Fixed knobs: seed=22; guidance=7.5; sampler=Euler; ControlNet start/end=0/1; upscaler=1.0.

Varied: `illusion_strength` $\in \{0.8, 1.5\}$ and prompt index `p1{p4}`.

Audio tool: HF Space *nsaintsever/music-generation* (model *facebook/musicgen-small*); duration=10s; use_sampling=True; top_k=250. (use_sampling and top_k are already fixed)

B. IDs & Prompts

Shorthand. `p1..p4` index the four prompts below; `s0p8` means `illusion_strength=0.8` and `s1p5` means 1.5.

Prompts

- `p1`: Forest at dawn break, trees on both sides, a black panther crouched on one of the trees, an owl on another tree. (neg: **Low quality**)
- `p2`: Forest at dawn, trees on both sides, a black panther on one tree, an owl on another. Optical illusion, hidden woman’s face, clear edges. (neg: **Low quality**)
- `p3`: Quiet forest at dawn, few trees only. Panther above, owl below. They align to form a hidden woman’s face. Optical illusion, clear edges, minimal background. (neg: **Multiple trees, multiple animals**)
- `p4`: Forest at dawn, panther on one tree, owl on another. They form a hidden woman’s face. Optical illusion, precise edges, gentle oil-paint look. (neg: **Low quality**)

File mapping `illusion_1.webp` — `p1 s0p8` (`q1-image-p1-s0p8`)

`illusion_2.webp` — `p1 s1p5` (`q1-image-p1-s1p5`)

`illusion_3.webp` — `p2 s0p8` (`q1-image-p2-s0p8`)

`illusion_4.webp` — `p2 s1p5` (`q1-image-p2-s1p5`)

`illusion_5.webp` — `p3 s1p5` (`q1-image-p3-s1p5`)

`illusion_6.webp` — `p3 s0p8` (`q1-image-p3-s0p8`)

`illusion_7.webp` — `p4 s0p8` (`q1-image-p4-s0p8`)

`illusion_8.webp` — `p4 s1p5` (`q1-image-p4-s1p5`)

C. Algorithm: IllusionScore computation (for Q3 image artifact)

Inputs. Control image *C* (`test.png`); generated image *G*.

Params (fixed). resize to 512×512 ; Canny with auto thresholds (median-based); 1 px edge dilation; EDP threshold = 0.10.

Outputs. Precision *P*, Recall *R*, F_2 , EDP, `portrait_pen`, IllusionScore.

1. **Preprocess.** Gray + resize both *C* and *G* to 512×512 .

2. **Edges.** Canny edges $\rightarrow E_C, E_G$; dilate both (1 px) for tolerance.
3. **Overlap.** $P = \frac{|E_G \wedge E_C|}{|E_G|}$, $R = \frac{|E_G \wedge E_C|}{|E_C|}$, $F_2 = \frac{5PR}{4P+R}$.
4. **Clutter penalty.** Edge density $d = \frac{|E_G|}{|G|}$; $EDP = \max(0, d - 0.10)$.
5. **Portrait penalty.** Frontal-face cascade on G ; let $a = \text{max face area} / \text{image area}$ (0 if none). Map a with a soft threshold near 0.08 to `portrait_pen` $\in [0, 1]$.
6. **Final score.** $\text{IllusionScore} = F_2 - 0.9\text{portrait_pen} - 0.25\text{EDP}$.

Intuition: the illusion succeeds when large contours in G cover control edges (high recall $\Rightarrow$ high F_2). Literal portraits and noisy textures are penalized.

Colab notebook: Metric.ipynb

D. Labeled exemplars (my three anchors)

Success: `illusion_2.webp` (q1-image-p1-s1p5) — $F_2=0.414$, `portrait`=0.000, `EDP`=0.110.

Borderline: `illusion_3.webp` (q1-image-p2-s0p8) — $F_2=0.292$, `portrait`=0.000, `EDP`=0.128.

Failure: `illusion_5.webp` (q1-image-p3-s1p5) — $F_2=0.281$, `portrait`=0.088, `EDP`=0.000.

E. How the prompt changed the score

- **Strength up (P1: 0.8 $\rightarrow$ 1.5).** F_2 rose (0.374 $\rightarrow$ 0.414) and `EDP` fell slightly (0.136 $\rightarrow$ 0.110) without triggering a face $\Rightarrow$ `illusion_2` outranks `illusion_1`.
- **“Hidden face, clear edges” (P2).** Helps alignment (F_2 0.292 $\rightarrow$ 0.377) and reduces clutter (`EDP` 0.128 $\rightarrow$ 0.025), but a small portrait penalty appears (0.099) at high strength $\Rightarrow$ overall score pulled down.
- **“Few trees, minimal background” (P3).** `EDP` ≈ 0 (clean) but recall is low ($F_2 \sim 0.28$): not enough structure to overlap the control; at 1.5 a face emerges $\Rightarrow$ portrait penalty.
- **“Precise edges, gentle oil-paint” (P4).** Oil-paint lowers `EDP` and often lifts F_2 , but smooth facial regions introduce a small portrait penalty; `illusion_8` wins because penalty stays modest.

F. Full evaluation table (all 8 images)

file	Illusion	F_2	Rec.	Prec.	EDP	Portrait	Notes
artifacts/samples/illusion_2.webp	0.386	0.414	0.614	0.180	0.110	0.000	P1, 1.5 (realistic)
artifacts/samples/illusion_8.webp	0.377	0.456	0.599	0.233	0.045	0.074	P4, 1.5 (oil-paint)
artifacts/samples/illusion_7.webp	0.370	0.484	0.693	0.219	0.084	0.103	P4, 0.8
artifacts/samples/illusion_1.webp	0.340	0.374	0.589	0.152	0.136	0.000	P1, 0.8
artifacts/samples/illusion_4.webp	0.282	0.377	0.470	0.210	0.025	0.099	P2, 1.5
artifacts/samples/illusion_3.webp	0.260	0.292	0.450	0.121	0.128	0.000	P2, 0.8
artifacts/samples/illusion_6.webp	0.216	0.287	0.271	0.377	0.000	0.078	P3, 0.8
artifacts/samples/illusion_5.webp	0.201	0.281	0.254	0.486	0.000	0.088	P3, 1.5

eval.csv schema (compact)

Column	Meaning (range/notes)
path	Relative filepath of the image scored.
eos_f2	Recall-weighted edge-overlap F_2 (0–1).
recall	Fraction of control edges hit (0–1).
precision	Fraction of predicted edges that match (0–1).
edp	Edge-density penalty = $\max(0, d - 0.10)$ (≥ 0).
portrait_pen	Soft face penalty from detector (0–1).
face_area_ratio	Largest face area / image area (0–1).
cascade_ok	Haar cascade loaded (True/False).
illusion_score	= eos_f2 – 0.9·portrait_pen – 0.25·edp.

G. Files included (for verification)

Control: artifacts/samples/test.png.

Images (8): artifacts/samples/illusion_1.webp ... illusion_8.webp.

Scores: eval.csv (all eight rows; not printed here).

Selected outputs (for grading):

```
{ "id": "q1-image-p1-s1p5", "file": "artifacts/samples/illusion_2.webp", "notes": "primary pick (realistic)" }
{ "id": "q1-image-p4-s1p5", "file": "artifacts/samples/illusion_8.webp", "notes": "primary pick (oil-paint)" }
{ "id": "q1-music-p6-d10", "file": "artifacts/samples/music_p6_d10.wav", "notes": "selected BGM" }
```

H. Image mosaic (all eight, in order)

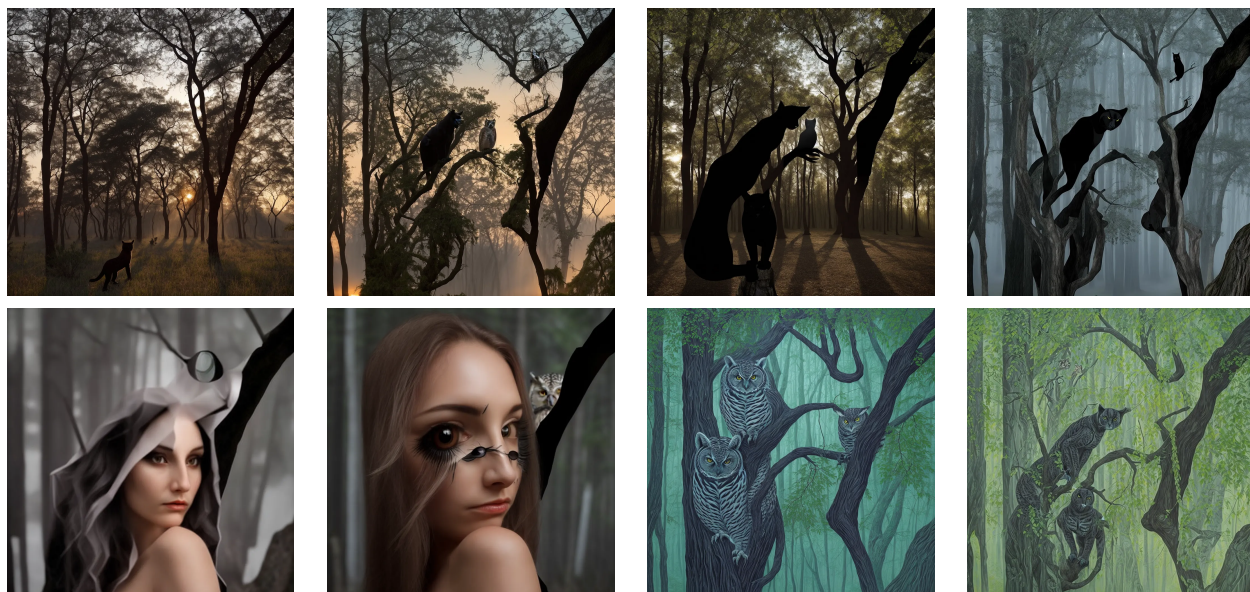


Figure 1: Eight generated images (1–8). See eval.csv for scores.